Certified Archimedean Stark Packets over Real Quadratic Fields: Orders Six and Ten

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Abstract

We give unconditional, replayable identifications of one-place archimedean Stark packets over real quadratic fields. Three proved theorem bases are used: quadratic analytic class-number formulae, Shintani's index-two algebraicity theorem followed by effective height rigidity, and descent to Stark's proved imaginary-quadratic rank-one theorem. The examples include, within a frozen literature perimeter, the first unconditional packets with character orders six and ten, an order-six packet at a ramified-prime-3 power conductor, a generic CM-descent packet of mixed signature, and one quartic packet proved through two disjoint theorem bases. A uniform quadratic-character formula and two structural lemmas explain the boundary of the three engines. In particular, a nontrivial one-place packet has even Shintani index. Every polynomial, Artin label, exponent, and analytic margin is tied to an exact or ball-arithmetic certificate. No unproved Stark conjecture and no floating-point recognition enters a proof.

1 Introduction and claim boundary

Let K be real quadratic and let $\mathfrak{m} = \mathfrak{f}\infty_2$ be a ray modulus containing one real place. For $A \in \mathrm{Cl}_{\mathfrak{m}}(K)$, let R be the sign class and put

$$Z_{\mathfrak{m}}(s, A) = \zeta_{\mathfrak{m}}(s, A) - \zeta_{\mathfrak{m}}(s, RA), \qquad X_A = \exp Z'_{\mathfrak{m}}(0, A). \tag{1}$$

The numbers X_A are positive real analytic invariants. Stark's rank-one conjecture predicts algebraic units with the prescribed Artin action, but that conjecture is not known in this generality. The purpose of this paper is to identify explicit packets only where a proved theorem already supplies algebraicity and where the remaining analytic-to-algebraic comparison can be certified.

The word *unconditional* has this precise meaning throughout: each displayed identity follows from a published proved theorem under hypotheses checked exactly in the case bundle. It does not mean that the general real-quadratic rank-one Stark conjecture is proved. Shintani's theorem is used only in its index-two range [1]; Stark's theorem is used only for abelian extensions of imaginary quadratic fields [2]; and the quadratic cases use the analytic class-number formula. Numerical PARI recognition, when retained as a cross-check, is outside the proof chain.

The historical "first" assertions are also bounded. We froze eleven sources comprising Shintani and Stark, Yamamoto's ray-invariant work, Tangedal's double-sine formulas, the Dummit–Sands–Tangedal computations and exponent-two theorem, Cohen–Roblot, Roblot, Kopp, and the archimedean/finite-place comparison literature [3, 4, 6, 7, 8, 9, 10, 11, 12]. Within that named and hash-frozen perimeter we found no earlier unconditional oriented packets with character order six or ten. This is a bounded literature statement, not a universal negative. In particular, the conditional or numerical constructions of Cohen–Roblot and Dummit–Sands–Tangedal are prior computations, not claims displaced by the present certification.

Main results

The packet order in the next theorem means the maximum order of a character occurring in the Fourier support of (1).

Theorem 1. *The following one-place archimedean packets are identified unconditionally.*

case	K	$\mathrm{N}\mathfrak{f}$	*support*	*safe exponent*	*margin*
RQ-000190	$\mathbb{Q}(\sqrt{7})$	*7*	$2, 6$	*4032*	> 5688
RQ-000419	$\mathbb{Q}(\sqrt{14})$	*7*	$2, 6$	*4032*	> 7315
RQ-000108	$\mathbb{Q}(\sqrt{5})$	*45*	4	*2880*	> 2460
RQ-000021	$\mathbb{Q}(\sqrt{2})$	*49*	$2, 6$	*2016*	> 4261
RQ-002057	$\mathbb{Q}(\sqrt{57})$	*27*	$2, 6$	*2592*	> 748
RQ-002955	$\mathbb{Q}(\sqrt{77})$	*7*	$2, 6$	*4032*	> 5151
RQ-001107	$\mathbb{Q}(\sqrt{33})$	*11*	$2, 10$	*15840*	> 5817

The first row is, within the frozen perimeter, the first unconditional order-six instance; the last row is the first order-ten instance. In addition:

1. *the primitive norm-*32 *packet over* $\mathbb{Q}(\sqrt{35})$*, and its norm-*64 *transport, are proved independently through* $\mathbb{Q}(\sqrt{-10})$ *and* $\mathbb{Q}(\sqrt{-14})$*;*
2. *three generic CM-descent packets, RQ-001569, RQ-007519, and RQ-001894, are proved independently through two imaginary quadratic bases each, with* $e = |\mu(E)| = 6$*;*
3. *the norm-*8 *packet RQ-000129 over* $\mathbb{Q}(\sqrt{6})$ *is proved through the* $e = 8$ *and* $e = 12$ *CM routes after an exact auxiliary-prime enlargement; and*
4. *the quartic packet RQ-000458 over* $\mathbb{Q}(\sqrt{14})$ *is proved once by Shintani's theorem and once by Stark's imaginary-quadratic theorem.*

The theorem concerns promoted case identities, not a census population. Counts, completeness, and conductor trends belong to a separate paper and are not logical inputs here.

2 Three proved engines

2.1 Quadratic characters

Suppose every character χ with $\chi(R) = -1$ has order two. Let χ° be primitive, let L_χ/K be its quadratic field, and let u_χ generate the primitive rank-one kernel of the exact norm map on free unit lattices. Orient u_χ to be larger than one at the selected real embedding and define

$$I_\chi = [E_{L_\chi}/\mu(L_\chi) : \langle E_K/\mu(K), u_\chi\rangle].$$

Proposition 2 (Uniform quadratic engine)**.** *With* E_χ *the exact product of imprimitive Euler factors at zero,*

$$L'_{\mathfrak{m}}(0, \chi) = E_\chi \frac{h_{L_\chi}}{h_K} \frac{w_K}{w_{L_\chi}} \frac{2}{I_\chi} \log |u_\chi|. \tag{2}$$

Moreover

$$Z'_{\mathfrak{m}}(0, A) = \frac{2}{|\operatorname{Cl}_{\mathfrak{m}}(K)|} \sum_{\chi(R)=-1} \chi(A)^{-1} L'_{\mathfrak{m}}(0, \chi). \tag{3}$$

Consequently a common denominator in (2)–(3) gives an explicit power identity for every X_A*.*

Proof. The field L_χ has signature $(2,1)$. Dividing the analytic class-number formulas for ζ_{L_χ} and ζ_K gives

$$L'_{\mathfrak{m}}(0,\chi) = E_\chi \frac{h_{L_\chi}}{h_K}\frac{w_K}{w_{L_\chi}}\frac{R_{L_\chi}}{R_K}.$$

In exact free-unit coordinates, let v_K be the embedded base unit and v_{rel} the primitive norm-kernel generator. Then $I_\chi = |\det(v_K, v_{\mathrm{rel}})|$. The regulator of the subgroup they generate is both $I_\chi R_{L_\chi}$ and $2R_K \log|u_\chi|$, proving (2). Equation (3) is exact Fourier inversion. All coefficients are rational because the supported characters are quadratic. $\square$

This proposition is a theorem, not a claim that every queued quadratic case has already been evaluated. Each application must still certify its conductor, class numbers, Euler factors, primitive norm kernel, determinant, and orientation.

2.2 Shintani algebraicity and height rigidity

Engine B first verifies Shintani's unit-congruence and real-place hypotheses and the index

$$[H : H \cap \mathbb{Q}^{\mathrm{ab}}] = 2. \tag{4}$$

The maximal absolutely abelian subfield is reconstructed from the commutator-fixed subgroup. Its imaginary quadratic bases are then reconstructed independently as ray fields; agreement of the two routes is mandatory.

For each divisor $\mathfrak{d}$ of the transfer conductor, the certificate prints the conductor factor, class number, roots of unity, real-distribution index, and a clearing exponent. The least common multiple is a safe exponent m. Shintani's theorem then makes X_A^m an algebraic unit with the required Artin action.

The analytic evaluator encloses all logarithms in Arb balls. If α_A is the isolated candidate and $\eta = X_A^m/\alpha_A$, these balls give an upper bound for $h(\eta)$. For degrees at least three, Voutier's explicit lower bound [5] excludes every non-torsion η. Degree one is handled by positivity of a rational unit; degree two by $\frac{1}{2}\log\phi$. Finally positivity eliminates the remaining root of unity. This is the height-rigidity step.

2.3 CM descent and Stark's proved theorem

Engine C starts from quartic ray-character support whose normal closure has the required V_4 projective quotient. It constructs the actual splitting closure, identifies both imaginary quadratic subfields, and checks linear reinduction on the normal closure. A finite exhaustive Dirichlet-coefficient signature separates the candidate character from its inverse; scalar twists are forbidden.

Let E/k be the selected abelian extension of an imaginary quadratic field. Stark's proved rank-one theorem applies with a set S containing the complex place and the conductor primes. In every use of the global-unit conclusion below, $|S| \geq 3$ is checked explicitly, and $e = |\mu(E)|$ is computed exactly. If $\ell_g = \log|g\varepsilon|_{\mathrm{ord}}$, the normalization is

$$\zeta'_S(0,g) = -\frac{2}{e}\ell_g, \qquad \ell_g = -\frac{e}{2}\zeta'_S(0,g). \tag{5}$$

For the frozen quartic Fourier convention,

$$L'_S(0,\psi) = -\frac{4}{e}(\ell_1 - i\ell_\sigma), \qquad \ell_1 - i\ell_\sigma = -\frac{e}{4}L'_S(0,\psi). \tag{6}$$

These are formulas, not interchangeable shorthand. The $e = 2$ anchor cannot distinguish $2/e$ from $e/2$; the nontrivial normalization check is the $e = 4$ cross-route identity for RQ-000458. The specializations of (5)–(6) carry the same minus signs for $e = 6, 8, 12$.

Arb lattice inversion isolates an integral orbit in $E^\times/\mu(E)$. The exact Artin bridge identifies its positive CM norms. The theorem therefore claims the torsion-invariant packet, not a literal root-of-unity representative.

3 The order-six and order-ten packets

3.1 $\mathbb{Q}(\sqrt{7})$ at the ramified prime above seven

For $K = \mathbb{Q}(\sqrt{7})$ and $\mathfrak{m} = \mathfrak{p}_7\infty_2$, the one-place and two-place ray groups are C_6 and $C_6 \times C_2$. The support orders are two and six and (4) holds. The commutator-fixed calculation produces $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{-7})$, and their intrinsic conductor reconstructions give the same degree-24 normal closure. The $\mathbb{Q}(i)$ route has two divisor rows and safe exponent 4032.

The packet polynomial over K is

$$\begin{aligned} P_7(X) = X^6 &- (6+2\sqrt{7})X^5 + (22+8\sqrt{7})X^4 \\ &- (34+13\sqrt{7})X^3 + (22+8\sqrt{7})X^2 \\ &- (6+2\sqrt{7})X + 1. \end{aligned} \tag{7}$$

It is irreducible over K, defines the one-place ray field, has six positive Sturm-isolated roots, and is Artin-labeled by the split primes 19 and 31.

For the class represented by $(3)^r$, an exact Yamamoto domain has generators

$$\lambda_r = (7-2\sqrt{7})/3^r, \qquad \lambda_r(8+3\sqrt{7}).$$

Its determinant is nine and its half-open parallelotope contains exactly nine lattice points. The resulting 27 independent double-sine evaluations are enclosed without a PARI L-value. At exponent 4032, the powered height upper bound is 9.1903×10^{-9}. The minimum Voutier lower bound in degrees three through 24 is 5.2279×10^{-5}, so the margin exceeds 5688. The degree-one and degree-two fallbacks finish the proof of (7).

3.2 Replications and the ramified-prime-3 control

For $\mathbb{Q}(\sqrt{14})$ at $\mathfrak{p}_7$, both $\mathbb{Q}(\sqrt{-7})$ and $\mathbb{Q}(\sqrt{-2})$ reconstruct the degree-24 normal closure. The latter route has clearing exponents 576 and 84, with least common multiple 4032. The packet is

$$\begin{aligned} P_{14}(X) = X^6 &- (13+4\sqrt{14})X^5 + (85+22\sqrt{14})X^4 \\ &- (139+38\sqrt{14})X^3 + (85+22\sqrt{14})X^2 \\ &- (13+4\sqrt{14})X + 1. \end{aligned} \tag{8}$$

Split primes 11 and 103 label the Artin generators. The 20-point cone evaluator gives powered height at most 7.1466×10^{-9}, hence margin greater than 7315.

The strongest boundary control is RQ-002057 over $\mathbb{Q}(\sqrt{57})$, with modulus of norm 27. The conductor is a power of the ramified prime 3 and the support still has order six. The bases $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(\sqrt{-19})$ independently give the same degree-24 closure. On the shorter route the divisor exponents are $864, 324, 108$, hence $m = 2592$. The fundamental positive unit has order three modulo the finite modulus, so a ray-class domain is the union of three adjacent cones, with 240 exact affine points per class. The powered height is at most 6.982×10^{-8}, giving margin greater than 748. Thus order-six support and ramified-prime-3 structure are both reachable; neither alone explains the separate $\mathbb{Q}(\sqrt{21})$ boundary.

3.3 Order ten

For RQ-001107 over $\mathbb{Q}(\sqrt{33})$ and $\mathfrak{p}_{11}$, put $y^2 - y - 8 = 0$. The packet is

$$\begin{aligned} P_{33}(X) = X^{10} &- (8+4y)X^9 + (74+30y)X^8 - (294+125y)X^7 \\ &+ (669+281y)X^6 - (871+368y)X^5 + (669+281y)X^4 \\ &- (294+125y)X^3 + (74+30y)X^2 - (8+4y)X + 1. \end{aligned} \tag{9}$$

Its absolute degree-20 field is the certified ray field over K. The actual packet-comparison degree is 40; the certificate conservatively covers degrees three through 80. On this whole window the minimum Voutier bound is

$$5.22795332222684\ldots \times 10^{-5}.$$

At $m = 15840$, the factor-100 raw-error target is $3.30047558221391\ldots \times 10^{-11}$; the achieved bound is 5.6731×10^{-13}, so the certified margin exceeds 5817. The degree-one and degree-two fallbacks are recorded separately.

4 CM-descent packets

4.1 A generic mixed-signature packet

The first packet completed by the generic Engine-C implementation is over $\mathbb{Q}(\sqrt{35})$, at the primitive ideal of norm 32. Both $\mathbb{Q}(\sqrt{-10})$ and $\mathbb{Q}(\sqrt{-14})$ have $e = 2$ and $|S| = 3$, and both give the Artin-labeled polynomial

$$\begin{aligned} P_{35}(X) = X^8 &- 38904X^7 + 905404X^6 + 62136X^5 \\ &- 873210X^4 + 62136X^3 + 905404X^2 - 38904X + 1. \end{aligned} \tag{10}$$

Exact conjugation fixes the four Artin norm classes, and Arb orbit isolation matches them to the four real roots. The remaining roots form two nonreal conjugate pairs, so the signature is $[4, 2]$. An exact conductor transport proves the norm-64 member.

The first $e = 6$ tranche comprises RQ-001569, RQ-007519, and RQ-001894. Six independent imaginary-base routes cover fourteen transported occurrences. Every route has $|S| \geq 3$, an exact character-selection signature, a certified theta/Mellin target, a unique integral anti-unit orbit, and an exact Artin bridge. The two routes for each canonical case give the identical degree-eight packet. Their complete integer polynomials are printed in the certificate bundle rather than abbreviated here.

4.2 $\mathbb{Q}(\sqrt{6})$ and auxiliary-prime independence

For RQ-000129 the two routes are over $\mathbb{Q}(\sqrt{-2})$, with $e = 8$, and $\mathbb{Q}(\sqrt{-3})$, with $e = 12$. Their exact candidate orbits obey 256 common-normal-closure identities

$$q_8^3 = q_{12}^2. \tag{11}$$

The natural second route has $|S| = 2$, so it cannot use the $|S| \geq 3$ global-unit clause. This first attempt was halted rather than promoted.

At the auxiliary rational primes 3 and 5, the complete Euler factors agree on the source and both CM routes, with nonzero derivative multipliers

$$P_3(1) = 1 + i, \qquad P_5(1) = 2. \tag{12}$$

Enlarging S gives size three, preserves rank one, and produces normalized Arb targets overlapping the same primitive $L'(0)$-ball. On the exact unit lattice, the prime 5 doubles the natural orbit and hence identifies the square of every positive packet norm. Positivity gives the unique square root. This proves the torsion-invariant packet with minimal polynomial

$$X^8 + 24X^7 + 732X^6 - 9304X^5 + 53958X^4 - 9304X^3 + 732X^2 + 24X + 1. \tag{13}$$

4.3 Two disjoint theorem bases

For RQ-000458, $K = \mathbb{Q}(\sqrt{14})$, the finite ideal has HNF $\left[\begin{smallmatrix}12 & 0\\ 0 & 6\end{smallmatrix}\right]$ and $\mathrm{Cl}_{\mathfrak{m}}(K) \simeq C_4 \times C_2$. Both routes identify

$$P_{458}(X) = X^4 - (20+6\sqrt{14})X^3 + (138+36\sqrt{14})X^2 - (20+6\sqrt{14})X + 1. \tag{14}$$

The Engine-B proof uses safe exponent 1152 and gives powered height 8.0791×10^{-9}, hence Voutier margin greater than 6470.

The independent Engine-C proof reinduces linearly over $\mathbb{Q}(\sqrt{-42})$ and $\mathbb{Q}(\sqrt{-3})$. On the first base, $|S| = 3$ and $e = 4$. The coefficient at $n = 7$ uniquely selects the ray character. Arb inversion isolates

$$(-8,4), \quad (0,-4), \quad (8,-4), \quad (0,4),$$

and 32 exact common-closure identities prove (14). Thus one theorem base is Shintani 1978 and the other is Stark 1980; they share no certificate intermediate. Because the second-route selection seal was not timestamped before execution, the process tag is `DUAL_ROUTED`, not the stronger `DUAL_PROVED`. This process caveat does not weaken either mathematical proof.

5 Structural boundary lemmas

Lemma 3 (No absolute-abelian fourth engine). *A nontrivial one-place differenced invariant over a real quadratic field is not governed by an absolutely abelian ray field.*

Proof. If the one-place field $H/\mathbb{Q}$ were abelian, the complex conjugations above the two real places of K would be conjugate and, in an abelian group, equal. Every absolute character would therefore have equal parity at the two real places. The Fourier support of (1), however, consists of characters odd at the retained place and even at the omitted place. A nonzero such character cannot descend from a Dirichlet character. □

Engine A is not an exception: it reduces individual quadratic characters by (2), but does not assert that the entire mixed-signature one-place ray field is absolutely abelian.

Lemma 4 (Index parity). *Let H/K be a one-place ray field and let $N/\mathbb{Q}$ be its genuine normal closure. If the sign class R is nontrivial, then*

$$2 \mid [H : H \cap \mathbb{Q}^{\mathrm{ab}}]. \tag{15}$$

Equivalently, an odd Shintani index can occur only when the Fourier support of (1) is empty.

Proof. Put $G = \mathrm{Gal}(N/\mathbb{Q})$, $A = \mathrm{Gal}(N/K)$, and $B = \mathrm{Gal}(N/H)$. Let $c_1, c_2 \in A$ be the real-place inertia involutions. A lift of the nontrivial automorphism of $K/\mathbb{Q}$ interchanges them, so they are conjugate in G. Hence $c_1c_2^{-1} \in [G,G]$. In $A/B = \mathrm{Gal}(H/K)$, the involution at the omitted place is trivial and the one at the retained place is R. Thus the image of $[G,G]$ in A/B contains R. Its order is $[H : H \cap \mathbb{Q}^{\mathrm{ab}}]$, which is even if $R \neq 1$. Finally, characters of a finite abelian group separate a nonidentity element of order two, so $R = 1$ is equivalent to empty differenced support. □

The use of the genuine normal closure in Lemma 4 is essential. Conjugation coordinates at an unstable modulus do not define $[G,G]$.

6 Reproducibility and containment

The exact field computations use PARI/GP 2.15.4; analytic certificates use Python 3.12.3 and python-flint 0.9.0. The principal replays were run under Linux/x86-64 on an AMD EPYC 9354P virtual CPU. All analytic conclusions are Arb balls, never point estimates.

The principal machine-readable records are

result	record
first order six	`data/q7-p7-case-v1.json`
second order six	`data/q14-p7-case-v1.json`
ramified-prime-3 control	`data/q57-norm27-case-v1.json`
order ten	`data/q33-p11-order10-case-v1.json`
generic CM packet	`artifacts/engine-c-w3-tranche-01-verified-v1.json`
$e = 6$ tranche	`artifacts/engine-c-e6-tranche-01-verified-v1.json`
$\mathbb{Q}(\sqrt{6})$	`data/q6-norm8-case-v2.json`
two-route packet	`data/rq000458-dual-case-v1.json`
quadratic theorem	`data/engine-a-uniform-theorem-v1.json`

Each record carries hashes of its exact and analytic dependencies. Failed attempts remain versioned. Predicate provenance is marked `GENUINE` or `PROXY`; no theorem tag may depend on a proxy. A later audit found an unmarked conjugation proxy in census classification, but none in a promoted theorem chain. The present paper was therefore frozen independently of all census populations and trends. This is a claim-boundary statement, not evidence for census completeness.

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